Telmo Cunha

https://tabdc.github.io/
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Research Experience

- **Research Assistant at Instituto Superior Técnico** — Lisbon, Portugal
  Ergodicity breaking in quantum many-body theory — *03/2026 - 10/2026*
  - Developed Python-based numerical simulations of constrained dynamical systems using Monte Carlo and exact diagonalization methods; analyzed large computational datasets to identify signatures of a phase transition.
- **Summer Research Assistant at Aalto University (AScI Program)** — Helsinki, Finland
  Equivariance properties of Graph Neural Networks (GNNs) — *04/2023 - 09/2023*
  - Developed analytical results concerning equivariance properties of Graph Neural Networks.
- **Summer Research Assistant at Instituto Gulbenkian de Ciência (IGC)** — Oeiras, Portugal
  Evolution of cancer cell populations — *06/2017 - 08/2017*
  - Developed and analysed an ODE mutation-selection model of cancer-cell population dynamics.

Teaching Experience

- **Invited Assistant Professor at ISCTE – Instituto Universitário de Lisboa** — Lisbon, Portugal
  Taught Calculus I, Calculus II and Linear Algebra to first-year Economics students, including classes taught in English. — *2024, 2025*
- **Private Tutor - Mathematics and Physics** — Lisbon, Portugal
  Individual instruction in mathematics and physics to high-school and undergraduate students. — *2015 – 2023*

Education

- **University of Lisbon - Instituto Superior Técnico (IST)** — Lisbon, Portugal
  MSc in Applied Mathematics and Computation (Final grade: 17/20) — *2025*
  Thesis topic: Analysis of Nonlinear Wave-Equations (Grade: 19/20)
- **University of Lisbon - Instituto Superior Técnico (IST)** — Lisbon, Portugal
  BSc in Engineering Physics (Final grade: 15/20) — *2019*
- **École Polytechnique Fédérale de Lausanne (EPFL)** — Lausanne, Switzerland
  Swiss-European mobility exchange program in theoretical physics — *2017*

Technical Skills

- **Programming**: Python (proficient), C/C++ and R (academic experience).
- **Python**: NumPy, SciPy, pandas, scikit-learn, PyTorch, matplotlib.
- **Mathematics/Computing**: Mathematical and numerical modelling, dynamical systems, optimization, machine learning, statistical/data analysis.
- **Tools**: Mathematica, LaTeX, Git.

Selected Projects

- **Mathematical Foundations of Machine Learning**: Wrote a set of notes covering the foundations of Learning Theory, Machine Learning and Kernel Methods with implementations in Python (repo).
- **Manifold Learning**: Applied manifold learning and nonlinear optimization techniques to dimensionality-reduction problems, with implementations in Python (repo).
- **Feature Selection**: Implemented information-theoretic feature-selection methods in Python and evaluated their effectiveness for dimensionality reduction and predictive modelling (repo).
- **Collaborative Filtering**: Developed a movie recommendation system using matrix-factorisation-based collaborative filtering, implementing model training and evaluation in Python (repo).

Additional Training

- **Quantum Landscapes International School**: Paris, France, 2026.
- **Aerospace Information Technology Summer School**: Würzburg, Germany, 2015.

Language Skills

- **Languages**: Portuguese (Native); English (Fluent); Spanish (Basic); French (Basic).